Shahab Ul Hassan

shahab.hasan94@gmail.com | +60-186657635 | Malaysia

LinkedIn | Google Scholar | Scopus | ResearchGate | Web of Science | ORCID | GitHub

Shahab Ul Hassan holds a PhD and MSc in Information Technology from Universiti Teknologi PETRONAS (UTP), Malaysia. He also holds a Bachelor's degree from the Institute of Business and Management Sciences, Peshawar, Pakistan. His research interests include machine learning, deep learning, computer vision, and explainable artificial intelligence (XAI), with a particular focus on healthcare applications. His work emphasizes medical image analysis, especially lung cancer detection and classification using advanced deep learning models. He is also actively involved in developing interpretable AI frameworks to improve the transparency, reliability, and clinical trustworthiness of AI-driven healthcare systems.

EDUCATION

10 Mar 2023 - 01 Aug 2026 (Expected)

Doctor of Philosophy in Information Technology

Universiti Teknologi PETRONAS, Perak (Malaysia)

Department: Computing | Study Mode: Full-time by Research

Thesis: Perceptually Weighted K-Median Segmentation with Adaptive Superpixel Perturbation-based LIME for Interpretable Computed Tomography Analysis.

15 Feb 2020 - 01 Sep 2022

Master of Science in Information Technology

Universiti Teknologi PETRONAS, Perak (Malaysia)

Department: Computing | Study Mode: Full-time by Research

Thesis: Classification of cardiac arrhythmia using hybrid convolutional neural network and bidirectional long short-term memory.

01 Sep 2012 - 30 Oct 2016

Bachelor of Science in Computer Science

Institute of Business and Management Sciences, (Pakistan)

Department: Computer Science | Study Mode: Full-time

FYP: Accident Detection and Rescue Application for Motorcycle.

EXPERIENCE

15 Feb 2020 - 06th Feb 2026

Teaching Assistant

Universiti Teknologi PETRONAS, Perak (Malaysia)

Achievements: Prepared teaching materials, tutorials, and evaluations; supervised students; invigilated examinations; marked assignments, quizzes, and tests. Taught courses over eleven semesters, including:

- Strategic Business Analytics (GFB3203)
- Software engineering & HCI (TEB2014/ TDB2163)
- Visual Programming (FBT0025)
- Structured Algorithm & Programming (FBT0015)
- Computer Systems (TEB1024)
- Cloud Computing (TFB2063)

15 Feb 2020 - Present

Research Assistant

Universiti Teknologi PETRONAS, Perak (Malaysia)

Achievements: Conducted and supported research; developed and evaluated experimental models; prepared manuscripts, technical reports, and grant proposals; presented at conferences; contributed to 5 peer-reviewed publications; participated in departmental academic activities.

Total Working Experience: Over 5 years of academic experience, including teaching and research.

Grants

1. Grant Title: Predicting Cardiovascular Diseases Using Interpretable Machine Learning Model and Smart IOT Platform. Yayasan Universiti Teknologi PETRONAS Fundamental Research (YUTP-FRG), Grant [015LCO-244].
2. Grant Title: Optimization of One-Dimensional Convolutional Neural Network for Corrosion Prediction in Subsea Pipelines. Institute of Autonomous System, Centre for Research in Data Science, Grant (YUTP-FRG), Cost Centre [015LC0-308].
3. Grant Title: Computer Aided Defect Detection of Limited Weld Inspection Films from Non-Destructive Radiography Testing. Institute of Autonomous System, Centre for Research in Data Science, Grant (YUTP-FRG), Cost Centre [015LC0-521].

PUBLICATIONS

Hassan, S. U*., Abdulkadir, S. J., Alhussian, H. S., Fayyaz, A. M., Al-Selwi, S. M., Khan, U., & Ismail, A. O. A. (2026). *Explainable deep learning-based lung cancer diagnosis using clinically-guided local interpretable model-agnostic explanations*. Scientific Reports. ([Q1, IF 3.9] [ISI, Scopus])

Hassan, S. U*., Abdulkadir, S. J., Zahid, M. S. M., & Al-Selwi, S. M. (2025). *Local Interpretable Model-Agnostic Explanation Approach for Medical Imaging Analysis: A Systematic Literature Review*. Computers in Biology and Medicine, 185, 109569. ([Q1, IF 6.3] [ISI, Scopus])

Hassan, S. U*., Mohd Zahid, M. S., Abdullah, T. A., & Husain, K. (2022). *Classification of cardiac arrhythmia using a convolutional neural network and bi-directional long short-term memory*. Digital health, 8, 20552076221102766. ([Q1, IF 3.3] [ISI, Scopus])

Hassan, S. U., Zahid, M. S. M., & Husain, K. (2020, October). *Performance comparison of CNN and LSTM algorithms for arrhythmia classification*. In 2020 International Conference on Computational Intelligence (ICCI) (pp. 223-228). IEEE. ([Scopus Index, IEEE Xplore])

Hassan, S. U*., Abdulkadir, S. J., Zahid, M. S. M., Fayyaz, A. M., Al-Selwi, S. M., & Sumiea, E. H. (2024). *An Optimized CNN-LSTM Model for Detecting Cardiac Arrhythmias*. In 2024 IEEE 8th International Conference on Signal and Image Processing Applications (ICSIPA) (pp. 1-6). IEEE. ([Scopus Index, IEEE Xplore])

Hassan, S. U*., Mohd Zahid, M. S., and Husain, K. *'Impact of bi-directional LSTM layer variation on cardiac arrhythmia detection performance'*. Published, 'The 5th international multi conference on artificial intelligence technology' (M-CAIT). **(Best Paper Award).** ([Scopus Index, IEEE Xplore])

Fayyaz, A. M., Abdulkadir, S. J., Talpur, N., Al-Selwi, S. M., Hassan, S. U., & Sumiea, E. H. (2025). *Grad-CAM (Gradient-weighted Class Activation Mapping): A systematic literature review*. *Computers in Biology and Medicine*, *198*, 111200. ([Q1, IF 6.3] [ISI, Scopus])

Fayyaz, A. M., Abdulkadir, S. J., Hassan, S. U., Al-Selwi, S. M., Sumiea, E. H., & Talib, L. F., & (2025). *The Role of Advanced Machine Learning in COVID-19 Medical Imaging: A Technical Review*. Results in Engineering. ([Q1, IF 7.9] [ISI, Scopus])

Abdullah, T. A., Zahid, M. S. M., Ali, W., & Hassan, S. U*. (2023). *B-LIME: An improvement of LIME for interpretable deep learning classification of cardiac arrhythmia from ECG signals*. Processes, 11(2), 595. ([Q1, IF 2.8] [ISI, Scopus])

Fayyaz, A. M., Abdulkadir, S. J., Al-Selwi, S. M., Sumiea, E. H., Iqbal, S., & Hassan, S. U. (2025). *A Novel CNN Model with Entropy-Coded Genetic Algorithm for Blood Cell Classification*. Journal of Advanced Research in Applied Sciences and Engineering Technology, 63(3). ([Q2, IF 1.71] [Scopus])

Husain, K., Mohd Zahid, M. S., Hassan, S. U., Hasbullah, S., & Mandala, S. (2021). Advances of ecg sensors from hardware, software and format interoperability perspectives. *Electronics*, *10*(2), 105. ([Q1, IF 2.6] [ISI, Scopus])

Daha, M. Y., Zahid, M. S. M., Alashhab, A., & Hassan, S. U.. (2021, December). Comparative analysis of community detection methods for link failure recovery in software defined networks. In *2021 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)* (pp. 157-162). IEEE.. ([Scopus Index, IEEE Xplore])

WORKSHOPS

26 Sep 2025 — **Conducting Systematic Literature Reviews: Hands-on Workshop for Researchers**
Department of Computing, Universiti Teknologi PETRONAS (Malaysia)

25 Jul 2023 — **LaTeX Workshop**
Department of Computing, Universiti Teknologi PETRONAS, Perak (Malaysia)

27 Oct 2023 — **Kill the Academic Noise**
Department of Computing, Universiti Teknologi PETRONAS, Perak (Malaysia)

Apr 2022 — **Mendeley: An Essential Tool for Researchers**
IEEE Universiti Teknologi PETRONAS, Perak (Malaysia)

SKILLS

Research & Documentation Tools
LaTeX, TeXstudio, EndNote, Mendeley, JabRef, MiKTeX, and MS Office.

Programming Languages and Techniques
Python, MATLAB, Java, JavaScript, SQL, Power BI, HTML, Jupyter Notebook, Simulink.

MISCELLANEOUS

04 Nov 2024 — **Graduate Technologist - Malaysia Board of Technologies (MBOT)**

08 Sep 2024 — **IEEE 8th International Conference on Signal and Image Processing Applications (ICSIPA)**
Concorde Hotel, Kuala Lumpur, (Malaysia)

20 July 2022 — **Si2TE Sirim invention, innovation and technology Exhibition (Bronz Medal)**
Sirim Kulim, Kedah, (Malaysia)

18 July 2021 **5th International Multi-Conference on Artificial Intelligence Technology (MCAIT)**
Universiti Kebangsaan Malaysia, Selangor (Malaysia)

08 Oct 2020 **International Conference on Computational Intelligence (ICCI)**
Universiti Teknologi PETRONAS, Perak (Malaysia)

7 May 2026 **Novelty in Thesis Writing**
Universiti Teknologi PETRONAS, Perak (Malaysia)

LANGUAGES

Urdu (Native).
English (Fluent)

REFEREES

Dr. Said Jadid Abdulkadir
Associate Professor and Data Science Cluster Head, Universiti Teknologi PETRONAS (Malaysia).
Email: saidjadid.a@utp.edu.my

Dr. Mohd Soperi Mohd Zahid
Assistant Professor, Faculty of Computing and Analytics Open University Malaysia (OUM), (Malaysia).
Email: soperi@oum.edu.my

Dr. Md Khorshed Alam
Staff Engineer (Data Scientist), OSRAM Opto Semiconductors (M) Sdn Bhd OS R&D STM APAC (Malaysia).
Email: md-khorshed.alam@ams-osram.com